

2015-2016 Spring
MAT124 Midterm
14/04/2016

1. a. Find the length of the parametric curve

$$x = (1 + \cos t) \cos t, \quad y = (1 + \cos t) \sin t, \quad 0 \leq t \leq 2\pi.$$

- b. Find parametric equations of the tangent line to the parametric curve $\vec{r} = t\vec{i} + t^2\vec{j} + t^3\vec{k}$, $-\infty < t < \infty$, at the point with $t = 2$.

2. Consider the functions $f(x, y) = x^2y$ and $g(x, y) = x^3 + 5y^2$.

- a. Compute ∇f and ∇g . ($\nabla = \text{grad}$)
- b. Find all unit vectors $\vec{u} = a\vec{i} + b\vec{j}$ along which the directional derivatives of f and g at the point $P_0(2, 1)$ are equal.
- c. Find all points $P(x, y)$ in the plane at which the directional derivatives of f and g in the direction of $\vec{v} = \vec{i} + \vec{j}$ are zero.

3. a. Show that the limit of $f(x, y)$ as (x, y) approaches to $(0, 0)$ along any line through the origin is the same.

- b. Show that $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

4. Find and classify the critical points of $f(x, y) = 2x^2 + y^2 + 2x^2y$.

5. Find the absolute maximum and minimum values of the function

$$f(x, y) = 2(x^2 + y^2 - 1)^2 + x^2 - y^2$$

on the unit disk $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$.

1. a. The length of a parametrized curve can be calculated with the formula

$$L = \int_a^b \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt.$$

Find the derivatives of x and y with respect to t .

$$\frac{dx}{dt} = -\sin t \cos t - (1 + \cos t) \sin t, \quad \frac{dy}{dt} = -\sin^2 t + (1 + \cos t) \cos t$$

Then

$$\left(\frac{dx}{dt}\right)^2 = \sin^2 t \cos^2 t + 2 \sin^2 t \cos t (1 + \cos t) + (1 + \cos t)^2 \sin^2 t,$$

$$\left(\frac{dy}{dt}\right)^2 = \sin^4 t - 2 \sin^2 t (1 + \cos t) \cos t + (1 + \cos t)^2 \cos^2 t.$$

The middle terms cancel each other in the addition. Recall the trigonometric identities $\sin^2 t + \cos^2 t = 1$ and $\cos 2t = 2 \cos^2 t - 1$. We have

$$\begin{aligned} \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= \sin^2 t \cos^2 t + (1 + \cos t)^2 \sin^2 t + \sin^4 t + (1 + \cos t)^2 \cos^2 t \\ &= \sin^2 t \cos^2 t + \sin^4 t + (1 + \cos t)^2 \sin^2 t + (1 + \cos t)^2 \cos^2 t \\ &= \sin^2 t (\sin^2 t + \cos^2 t) + (1 + \cos t)^2 (\cos^2 t + \sin^2 t) \\ &= \sin^2 t + (1 + \cos t)^2 \\ &= \sin^2 t + 1 + 2 \cos t + \cos^2 t = 2 + 2 \cos t = 4 \cos^2 \frac{t}{2}. \end{aligned}$$

The length of the curve is

$$L = \int_0^{2\pi} \sqrt{4 \cos^2 \frac{t}{2}} dt = \int_0^{2\pi} 2 \left| \cos \frac{t}{2} \right| dt = 2 \cdot 2 \int_0^{\pi} \cos \frac{t}{2} dt = 4 \cdot \frac{\sin \frac{t}{2}}{\frac{1}{2}} \Big|_0^{\pi} = \boxed{8}.$$

- b. Find the derivative of the parametric equation with respect to t .

$$\frac{d\vec{r}}{dt} = \frac{d}{dt}(t) \vec{i} + \frac{d}{dt}(t^2) \vec{j} + \frac{d}{dt}(t^3) \vec{k} = \vec{i} + 2t \vec{j} + 3t^2 \vec{k}$$

Evaluate $\vec{r}$ and $\vec{r}'$ at $t = 2$.

$$\vec{r}|_{t=2} = \vec{i} + 4\vec{j} + 8\vec{k}, \quad \vec{r}'|_{t=2} = 2\vec{i} + 4\vec{j} + 8\vec{k}$$

The parametric equations for a line that passes through the point $P_0(x_0, y_0, z_0)$ is

given by

$$\left. \begin{aligned} x &= r_1 + r'_1 t \\ y &= r_2 + r'_2 t \\ z &= r_3 + r'_3 t \end{aligned} \right\} \quad t \in \mathbb{R}.$$

Therefore, the parametric equations for the line L is

$$\left. \begin{aligned} x &= 2 + t \\ y &= 4 + 4t \\ z &= 8 + 12t \end{aligned} \right\} \quad t \in \mathbb{R}.$$

2. a. The gradient of each function is below.

$$\begin{aligned} \nabla f &= \langle f_x, f_y \rangle = \langle 2xy, x^2 \rangle \\ \nabla g &= \langle g_x, g_y \rangle = \langle 3x^2, 10y \rangle \end{aligned}$$

b. Since $\vec{u}$ is a unit vector, we have $|\vec{u}| = \sqrt{a^2 + b^2} = 1$. The directional derivatives along the unit vector $\vec{u}$ evaluated at P_0 are

$$D_{\vec{u}}f(P_0) = \nabla f \cdot \vec{u}|_{P_0} = \langle 4, 4 \rangle \cdot \langle a, b \rangle = 4a + 4b,$$

$$D_{\vec{u}}g(P_0) = \nabla g \cdot \vec{u}|_{P_0} = \langle 12, 10 \rangle \cdot \langle a, b \rangle = 12a + 10b.$$

We have $D_{\vec{u}}f(P_0) = D_{\vec{u}}g(P_0)$.

$$4a + 4b = 12a + 10b \implies b = -\frac{4a}{3}$$

Along with $\sqrt{a^2 + b^2} = 1$, we find

$$\sqrt{a^2 + b^2} = 1 \implies a^2 + \left(\frac{-4a}{3}\right)^2 = 1 \implies 25a^2 = 9 \implies a = \pm \frac{3}{5}.$$

Hence

$$\vec{u} = \frac{3}{5}\vec{i} - \frac{4}{5}\vec{j} \quad \text{or} \quad \vec{u} = -\frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}.$$

c. Since $\vec{v}$ is not a unit vector, define the unit vector $\vec{w} = \frac{\vec{v}}{|\vec{v}|} = \frac{\vec{i} + \vec{j}}{\sqrt{2}}$.

Set the directional derivatives to 0.

$$D_{\vec{w}}f = 0 \implies \nabla f \cdot \vec{w} = 0 \implies \langle 2xy, x^2 \rangle \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = 0 \implies 2xy + x^2 = 0$$

$$D_{\vec{w}}g = 0 \implies \nabla g \cdot \vec{w} = 0 \implies \langle 3x^2, 10y \rangle \cdot \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = 0 \implies 3x^2 + 10y = 0$$

Factor x in the first equation.

$$x(2y + x) = 0 \implies x = 0 \text{ or } x = -2y$$

Consider the second equation with the relations found above.

$$3x^2 + 10y = 0 \implies \begin{cases} x = 0 \implies y = 0 \\ x = -2y \implies y(12y + 10) = 1 \implies y = 0 \text{ or } y = -\frac{10}{12} \end{cases}$$

All the points are $(x, y) = (0, 0)$ and $(x, y) = \left(\frac{5}{3}, -\frac{5}{6}\right)$.

3. a. Along any line through the origin (except the vertical line) can be defined by $y = mx$. Along $y = mx$,

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{x \rightarrow 0} f(x, mx) = \lim_{x \rightarrow 0} \frac{x^3(m^4x^4)}{x^4 + x^3(m^2x^2) + m^{10}x^{10}} \\ &= \lim_{x \rightarrow 0} \frac{m^4x^3}{1 + m^2x + m^{10}x^6} = \frac{0}{1} = \boxed{0}. \end{aligned}$$

Check the vertical line. Along $x = 0$,

$$f(0, y) = 0 \implies \lim_{(x,y) \rightarrow (0,0)} f(x, y) = \boxed{0}.$$

- b. We earlier showed that the limit along any line through the origin is 0. Apply the Two-Path test. Try $x = -y^2$.

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{y \rightarrow 0} f(-y^2, y) = \lim_{y \rightarrow 0} \frac{-y^6y^4}{y^8 - y^6y^2 - y^{10}} = \lim_{y \rightarrow 0} \frac{-y^{10}}{y^{10}} = -1 \neq 0$$

Since we have two different limits, the limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

4. Obtain the partial derivatives and set to 0.

$$f_x = 4x + 4xy = 0 \implies 4x(1 + y) = 0 \implies x = 0 \text{ or } y = -1$$

$$f_y = 2y + 2x^2 = 0 \implies \begin{cases} x = 0 \implies y = 0 \\ y = -1 \implies x = \pm 1 \end{cases}$$

Thus the critical points are $(0, 0)$, $(1, -1)$ and $(-1, -1)$. Obtain the Hessian determinant.

$$f_{xx} = 4 + 4y, \quad f_{yy} = 2, \quad f_{xy} = f_{yx} = 4x$$

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2 = 8 + 8y - 16x^2$$

Evaluate D at the points.

$$D(0, 0) = 8 > 0, \quad f_{yy}(0, 0) = 2 > 0 \implies (0, 0) \text{ is a local minimum.}$$

$$D(1, -1) = -16 < 0 \implies (0, 0) \text{ is a saddle point.}$$

$$D(-1, -1) = -16 < 0 \implies (0, 0) \text{ is a saddle point.}$$

$$\boxed{(0, 0) \text{ is a local minimum. } (1, -1) \text{ and } (-1, -1) \text{ are saddle points.}}$$

5. The extreme value theorem states that a continuous function on a specific region will always reach its absolute highest and lowest values.

On the interior of D ,

$$f_x = 4(x^2 + y^2 - 1)2x + 2x = 2x [4(x^2 + y^2) - 3] = 0,$$

$$f_y = 4(x^2 + y^2 - 1)2y - 2y = 2y [4(x^2 + y^2) - 5] = 0.$$

From the first equality, we have $x = 0$ or $x^2 + y^2 = \frac{3}{4}$.

From the second one, if $x = 0$, then $y = 0$ or $y = \pm \frac{\sqrt{5}}{2} \notin D$.

If $x^2 + y^2 = \frac{3}{4}$, then $y = 0$. Therefore, $x = \pm \frac{\sqrt{3}}{2}$.

So the critical points are $(0, 0)$, $(\frac{\sqrt{3}}{2}, 0)$ and $(-\frac{\sqrt{3}}{2}, 0)$.

On the boundary of D ,

$$x^2 + y^2 = 1 \implies y^2 = 1 - x^2 \implies y = \pm \sqrt{1 - x^2},$$

$$f(x, \pm \sqrt{1 - x^2}) = 2(x^2 + 1 - x^2 - 1)^2 + x^2 - 1 + x^2 = 2x^2 - 1.$$

Find the critical points on the boundary.

$$f'(x, \pm \sqrt{1 - x^2}) = 4x = 0$$

At $x = 0$, $y = \pm 1$. The critical points are $(0, 1)$ and $(0, -1)$.

The end points are $(1, 0)$ and $(-1, 0)$.

Evaluate f at all the aforementioned points and compare the values.

$$f(0, 0) = 2, \quad f(0, 1) = f(0, -1) = -1$$

$$f(1, 0) = f(-1, 0) = 1, \quad f\left(\frac{\sqrt{3}}{2}, 0\right) = f\left(-\frac{\sqrt{3}}{2}, 0\right) = 7/8$$